$$= 1-\{P(A)+P(B)-P(A\cap B)\}$$

$$= 1-P(A)-P(B)+P(A)\cdot P(B)$$

$$= (1-P(A))(1-P(B))$$

$$= P(\overline{A})\cdot P(\overline{B})$$

Thus $\overline{A}$ and $\overline{B}$ are independent .

Similarly one can prove (ii) and (iii).

Example 12.18

Two cards are drawn from a pack of 52 cards in succession. Find the probability that both are Jack when the first drawn card is (i) replaced (ii) not replaced

Solution

Let A be the event of drawing a Jack in the first draw,

B be the event of drawing a Jack in the second draw.

Case (i)

Card is replaced

$$n(A) = 4 \quad \text{(Jack)}$$
$$n(B) = 4 \quad \text{(Jack)}$$
$$\text{and} \quad n(S) = 52 \quad \text{(Total)}$$

Clearly the event A will not affect the probability of the occurrence of event B and therefore A and B are independent.

$$P(A\cap B) = P(A)\,.\,P(B)$$

$$P(A) = \frac{4}{52}\,,\ P(B)=\frac{4}{52}$$

$$P(A\cap B) = P(A)\ P(B)$$

$$= \frac{4}{52}\ .\ \frac{4}{52}$$

$$= \frac{1}{169}\ .$$

Case (ii)

Card is not replaced

In the first draw, there are 4 Jacks and 52 cards in total. Since the Jack, drawn at the first draw is not replaced, in the second draw there are only 3 Jacks and 51 cards in total. Therefore the first event A affects the probability of the occurrence of the second event B.

Thus A and B are not independent. That is, they are dependent events.

$$\text{Therefore, } P(A\cap B) = P(A)\,.\,P(B/A)$$

$$P(A) = \frac{4}{52}$$

12_11th-Maths-Vol2-Unit12.indd 254 23-03-2019 10:17:47